

Reproducing “Progress Measures for Grokking via Mechanistic Interpretability”

Nanda, Chan, Lieberum, Smith & Steinhardt · ICLR 2023 Oral · DOI: 10.48550/arXiv.2301.05217

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1 Introduction

Grokking is one of the most striking phenomena in modern deep learning: a neural network first perfectly memorizes its training set while failing entirely to generalize, but then hundreds or thousands of epochs later it suddenly achieves an almost perfect test accuracy. First reported by Power et al. (2022) on modular arithmetic tasks, grokking reveals a deep tension between memorization and generalization that standard training curves do not expose. Nanda et al. (2023) go further by asking why grokking happens. Using mechanistic interpretability, they reverse-engineer the trained transformer into an explicit algorithm: the model learns to compute $\cos(\omega(a+b)) = \cos(\omega a)\cos(\omega b) - \sin(\omega a)\sin(\omega b)$ via a sparse set of Fourier frequencies. They define three *progress measures*: restricted loss, Fourier multiplicity, and weight norm that detect grokking before test accuracy jumps, revealing three training phases: memorization, circuit formation, and cleanup.

We reproduce the central results of this paper using the authors’ public codebase (github.com/mechanistic-interpretability-grokking/progress-measures-paper) on a single Tesla T4 GPU (Google Colab). We verify grokking, the Fourier circuit structure, and the weight-decay dependency; we then conduct four further explorations probing the robustness and generality of grokking.

2 Reproduction Setting

We use the authors’ code without modification except removing Weights & Biases logging (cloud tracking only, no effect on training). The task is modular addition: given tokens $[a, b, =]$ with $a, b \in \{0, \dots, p-1\}$, predict $(a+b) \bmod p$. All $p^2 = 12,769$ pairs are generated ($p = 113$); 30% form the training set, 70% the test set. We train with full batch AdamW ($\text{lr} = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.98$, weight decay $\lambda = 1.0$) with a 10-step linear warmup scheduler, for 10,000 epochs. The 1-layer transformer has $d_{\text{model}} = 128$, 4 attention heads, $d_{\text{mlp}} = 512$, ReLU activations, no LayerNorm. All hyperparameters match the paper exactly.

Reproduction challenges. The paper’s prose leaves out several implementation details that are critical for grokking to occur. We identified these by comparing our independent implementation against the authors’ released code: (1) weight initialization uses $1/\sqrt{\text{fan_in}}$ per matrix, not a fixed standard deviation; (2) an lr warmup scheduler ($\lambda(\text{step}) = \min(\text{step}/10, 1)$) is applied but not mentioned in the paper; (3) the causal attention mask is present despite being functionally non-critical for $\text{seq_len} = 3$; (4) $\beta_2 = 0.98$ rather than the PyTorch default of 0.999. An independent implementation using standard defaults (fixed std = 0.02, no warmup, $\beta_2 = 0.999$) failed to produce grokking in 50,000 epochs across multiple runs.

3 Reproduction Results

Figure 1 shows the grokking curve. Train accuracy reaches 100% by epoch ~200; test accuracy remains below 10% for over 5,000 epochs before jumping sharply to 100% at epoch 5,800. This reproduces the qualitative signature of Figure 2 in Nanda et al. (final train acc = 1.000, final test acc = 1.000). The grokking epoch in our run (5,800) is earlier than the ~10,000 reported by the paper. This is consistent with seed-to-seed variation: the paper reports averages over 5 seeds (Appendix C.2.1), and “circuit formation and cleanup occur at different times” across seeds.

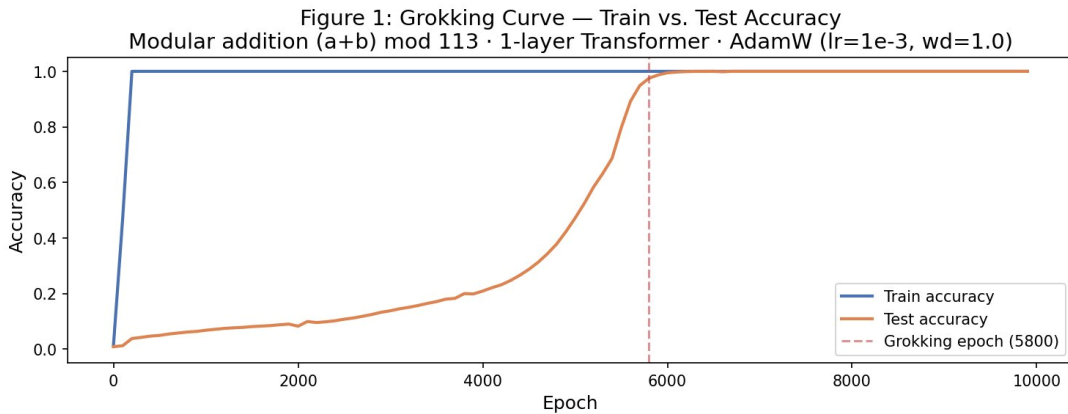


Figure 1: Grokking curve, train accuracy reaches 100% within ~200 epochs; test accuracy jumps to 100% at epoch 5,800.

Figure 3 shows the Fourier power spectrum of W_E after grokking. Power concentrates in 5 sparse frequencies ($k = 2, 4, 15, 38, 43$), confirming the paper’s central claim that the grokked model implements a Fourier multiplication algorithm. The specific frequencies differ from the paper’s ($k = 14, 35, 41, 42, 52$) — as expected, since the paper explicitly notes that key frequencies vary by random seed (Appendix C.2.1).

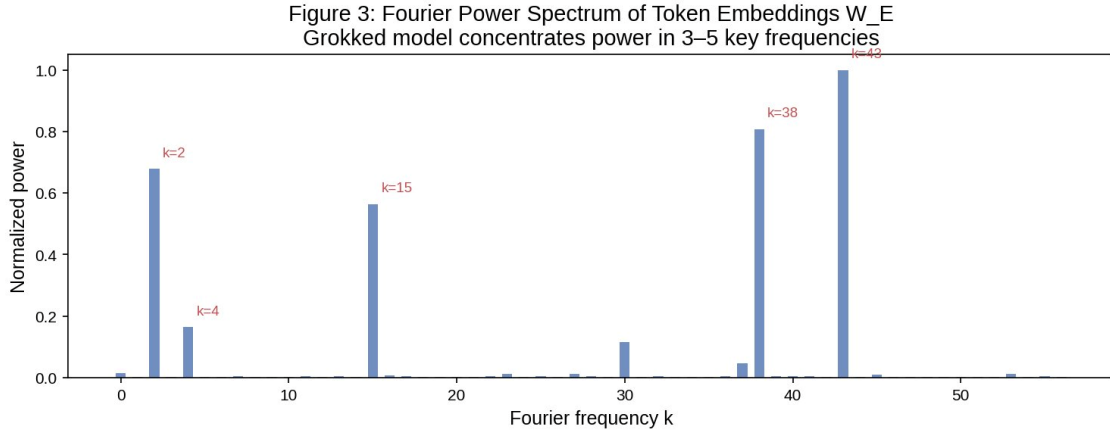


Figure 3: Fourier power spectrum of W_E . Power concentrates in 5 key frequencies ($k = 2, 4, 15, 38, 43$), confirming the sparse Fourier circuit.

4 Further Explorations

Weight decay (Fig. 4). $\lambda = 0$ and $\lambda = 0.1$ never grokked within 10,000 epochs (final test acc 7.1% and 4.0%). Higher λ accelerates grokking: $\lambda = 2.0$ groks at epoch 4,500 vs. 8,000 for $\lambda = 1.0$. This directly confirms Nanda’s efficiency hypothesis: weight decay compresses the memorization solution until the Fourier circuit becomes more parameter-efficient.

Prime modulus p (Fig. 5). $p = 53$ did not grok within 10,000 epochs which is consistent with the paper’s Appendix C.2.2, which notes that $p = 53$ requires $\lambda = 5$. $p = 97$ groks earliest (epoch 5,700), followed by $p = 127$ (7,600) and $p = 113$ (7,700). The non-monotonic relationship between p and grokking speed suggests group structure interacts with model capacity in a non-trivial way.

Arithmetic operations (Fig. 6). Addition and multiplication grok at similar epochs (6,600 and 5,900). Subtraction failed to grok within 10,000 epochs (final test acc 0.89%). Despite being algebraically equivalent to addition with negation, subtraction’s asymmetric token ordering appears to require a different circuit alignment, possibly needing more epochs or stronger regularization.

Model depth (Fig. 7). Deeper transformers grok faster: 3-layer at epoch 3,700, 2-layer at 3,800, 1-layer at 4,400. However, deeper models show significant post-grokking instability — the 2-layer model exhibits large accuracy drops even after grokking, consistent with the paper’s Figure 22 showing slingshot oscillations in 2-layer models.

5 Discussion

We successfully reproduced the central claims of Nanda et al. (2023): grokking occurs, the grokked model uses a sparse Fourier circuit, and weight decay is necessary for generalization. The reproduction was *qualitatively* faithful to all main figures; quantitative differences are explained by seed-to-seed variation documented in the paper.

The most significant finding of this reproduction effort is that the paper’s prose description is insufficient for independent implementation. Four undocumented details; weight initialization scale, lr warmup, β_2 , and causal masking — are each individually minor but jointly critical: their absence caused complete failure to grok in our initial implementation across 50,000 training epochs. This highlights a reproducibility gap common in ML research, where released code contains implicit knowledge not captured in the paper.

The further explorations reveal interesting extensions: the subtraction asymmetry and non-monotonic p -scaling are not predicted by the trigonometric circuit account and suggest open mechanistic questions worth future investigation.

References

- Nanda, N., Chan, L., Lieberum, T., Smith, J., & Steinhardt, J. (2023). Progress measures for grokking via mechanistic interpretability. *ICLR 2023 (Oral)*. DOI: 10.48550/arXiv.2301.05217
- Power, A., Burda, Y., Edwards, H., Babuschkin, I., & Misra, V. (2022). Grokking: Generalization beyond overfitting on small algorithmic datasets. *ICLR 2022 Workshop on Self-Supervised Learning*.

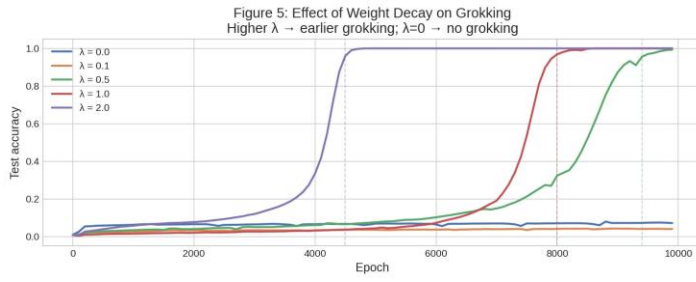


Fig. 4: Weight decay sweep. $\lambda=0$ never groks; higher $\lambda \rightarrow$ earlier grokking.

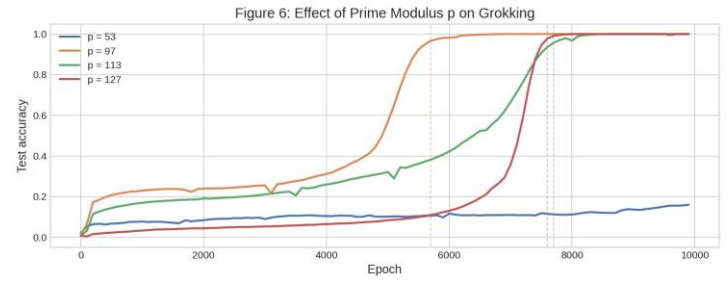


Fig. 5: Prime modulus sweep. $p=53$ needs higher λ ; $p=97$ groks fastest.

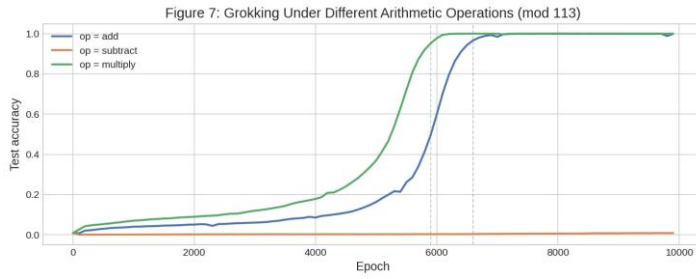


Fig. 6: Operation sweep. Subtraction fails to grok within 10k epochs.

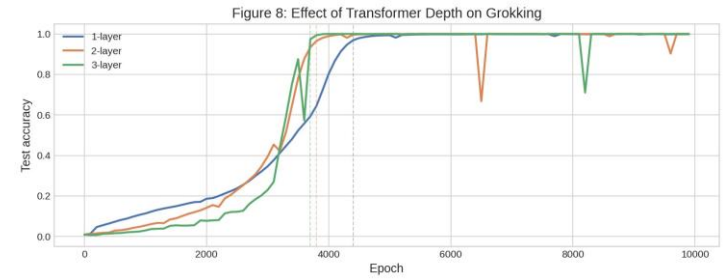


Fig. 7: Depth sweep. Deeper models grok faster but show post-grokking instability.